cse 242 hw2

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1

1. (a)

$$
\begin{aligned}
J(w) &= \Big|Xw - y\Big|_2^2 + \lambda\Big|w\Big|_2^2 \\
J(w) &= (Xw-y)^T(Xw-y) + \lambda w^T w \\
J(w) &= w^T X^T X w - w^T X^T y - y^T X w - y^T y + \lambda w^T w \\
J(w) &= w^T X^T X w - 2y^T X w - y^T y + \lambda w^T w \\
\frac{\partial J(w)}{\partial w} &= 2X^T X w - 2y^T X + 2\lambda w = 0
\end{aligned} \tag{1}
$$

then we see

$$
\begin{aligned}
2X^T X w + 2\lambda w &= 2y^T X \\
(X^T X + \lambda I)w &= y^T X
\end{aligned} \tag{2}
$$

then we get the solution in the hw.

(b) as $\lambda \to 0$, $w_{ridge} -> (X^T X)^{-1} X^T y$. as $\lambda \to \infty$, $w_{ridge} -> 0$. the former is when penalty goes to zero and the latter is when error dominates.

(c)

$$
\begin{aligned}
\Big|\hat{X}w - \hat{y}\Big| &= (\hat{X}w - \hat{y})^T(\hat{X}w - \hat{y}) \\
&= w^T \hat{X}^T \hat{X} w - \hat{y}^T \hat{X} w - w^T \hat{X}^T \hat{y} + \Big|\hat{y}\Big|^2 \\
&= w^T (X^T X + \sqrt{\lambda} X^T I_d + \sqrt{\lambda} I_d^T X + \lambda I) w \\
&\quad - \hat{y}^T \hat{X} w - w^T \hat{X}^T \hat{y} + \Big|\hat{y}\Big|^2
\end{aligned} \tag{3}
$$

note $(\sqrt{\lambda} I w)^2 = \lambda \sum w_i^2$. factoring out w^T, excluding the $w = 0$ solution, the argmin when zero gives the original expression.

2

(a)

$$\begin{aligned}(X^TX+\lambda I)^{-1} &= (Q\Lambda^2Q^T+\lambda I)^{-1} \\ &= (Q\Lambda Q^T+\lambda QIQ^T)^{-1} \\ &= (Q(\Lambda^2+\lambda I)Q^T)^{-1} \\ &= (Q^T)^{-1}((\Lambda^2+\lambda I))^{-1}Q^{-1} \\ &= Q((\Lambda^2+\lambda I))^{-1}Q^T\end{aligned} \tag{4}$$

λ inversely scales the eigenvalue

(b)

$$\kappa(X^TX) = \frac{\sigma_{max}^2}{\sigma_m^2 in} \tag{5}$$

since sigmas are positive, the inequality on the right holds.

the equality on the left is true because of part a. the eigenvalues are exactly $\Lambda + \lambda I$

we can also see why equality only hods when λ is zero, since every eigenvalue is different on the LHS when λ is nonzero.

(c) if there are zero eigenvalues, it means the matrix is invertable. the smallest eigenvalues with regularization are λ and no eigenvalues are zero, which implies full rank, which implies invertability.